

BATCH No:MAI195

**DEEP LEARNING-BASED DERMATOLOGICAL
DISEASE DETECTION USING CNN AND IMAGE
PROCESSING**

*Major project report submitted
in partial fulfillment of the requirement for award of the degree of*

**Bachelor of Technology
in
Computer Science & Engineering**

By

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*Under the guidance of
Dr. S. Selvin Ebenezer, M.E. Ph.D.,
ASSISTANT PROFESSOR*



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF
SCIENCE AND TECHNOLOGY**

(Deemed to be University Estd u/s 3 of UGC Act, 1956)

**Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA**

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CERTIFICATE

It is certified that the work contained in the project report titled "DEEP LEARNING-BASED DERMATOLOGICAL DISEASE DETECTION USING CNN AND IMAGE PROCESSING" by "CH. LAKSHMAN (21UECS0012), T. DHANUNJAY (21UECS0635), B. REVANTH (21UECM0029)" has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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May, 2025

DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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APPROVAL SHEET

This project report entitled "DEEP LEARNING-BASED DERMATOLOGICAL DISEASE DETECTION USING CNN AND IMAGE PROCESSING" by CH. LAKSHMAN (21UECS0012), T. DHANUNJAY (21UECS0635), B. REVANTH (21UECM0029) is approved for the degree of B.Tech in Computer Science & Engineering.

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ABSTRACT

Artificial Neural Networks are a special type of machine learning algorithms that are modeled after the human brain. That is, just like how the neurons in our nervous system are able to learn from the past data, similarly, the ANN is able to learn from the data and provide responses in the form of predictions or classifications. ANNs are nonlinear statistical models which display a complex relationship between the inputs and outputs to discover a new pattern. A variety of tasks such as image recognition, speech recognition, machine translation as well as medical diagnosis makes use of these artificial neural networks. An important advantage of ANN is the fact that it learns from the example data sets. Most commonly usage of ANN is that of a random function approximation. With these types of tools, one can have a cost-effective method of arriving at the solutions that define the distribution. ANN is also capable of taking sample data rather than the entire dataset to provide the output result. With ANNs, one can enhance existing data analysis techniques owing to their advanced predictive capabilities.

Keywords:

Skin Disease Classification, Convolutional Neural Network (CNN), Deep Learning, Medical Image Analysis, Streamlit Web App, PIL (Python Imaging Library).

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LIST OF ACRONYMS AND ABBREVIATIONS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
IP	Image Processing
ML	Machine Learning
SVM	Support Vector Machine

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Chapter 1

INTRODUCTION

1.1 Introduction

Skin conditions are prevalent global health concerns, impacting millions of individuals annually. Although most of these conditions are non-threatening, certain ones, like melanoma, can become fatal if not identified and treated in a timely manner.. Traditional methods for diagnosing skin conditions rely on visual inspection by healthcare professionals, often aided by dermatoscopic tools. However, the accuracy of diagnosis can be influenced by factors such as experience, the availability of medical professionals, and the complexity of certain skin lesionsRecent developments in artificial intelligence (AI) and machine learning (ML) have created new possibilities for automating and refining the assessment process, particularly in dermatology. Convolutional Neural Networks (CNNs), a specialized type of artificial neural network, have demonstrated outstanding performance in visual recognition tasks. Their ability to autonomously learn spatial patterns and structures from raw data has made CNNs the preferred method for medical image classification, including the detection of skin conditions.CNNs can be developed using large datasets of labeled images, enabling them to recognize patterns and features that may pose a challenge for detection by the human eye.. These models can assist in identifying a wide range of skin conditions, from common issues like acne and eczema to more severe diseases such as basal cell carcinoma, melanoma, and psoriasis. By providing accurate and rapid analysis, CNNs can assist healthcare providers in early diagnosis, potentially improving patient outcomes and reducing the burden on medical professionals.This paper investigates the application of CNNs for the automated identification and classification of skin conditions. We discuss the benefits of using CNNs in dermatology, the challenges associated with image preprocessing, dataset requirements, and the model architectures commonly used for skin condition classification.

1.2 Background

Dermatological conditions include a broad spectrum of skin diseases, such as infections, inflammatory conditions, and cancers, that afflict a large percentage of the world's population. Proper and timely diagnosis is important to provide the right treatment and avoid complications or disease progression. Nevertheless, traditional diagnostic techniques rely heavily on the skill and subjective experience of dermatologists, which may result in inconsistency in diagnosis, particularly in resource-poor or remote areas.

Over the past few years, advances in artificial intelligence (AI)[11], specifically deep learning, have brought with them new possibilities in medical image analysis. Convolutional Neural Networks (CNNs)[7], a type of deep learning model, have shown outstanding results on various image classification and pattern identification tasks. With their capability to extract and learn features from unprocessed image data autonomously, they are especially well suited for the detection of dermatological diseases.

When coupled with image processing methodologies like filtering, enhancement, and segmentation, CNN-based systems can enhance diagnostic precision and effectiveness. Such integration promises high potential in the creation of trustworthy, automatic tools to aid dermatologists, decrease workload, and offer accessible health-care solutions to wider populations

1.3 Objective

The main goal of this project is to create an intelligent, autonomous system capable of detecting and classifying dermatological diseases accurately with the help of deep learning, i.e., Convolutional Neural Networks (CNNs)[7]. With the help of a strong dataset of labeled skin images, the system should be able to detect several skin conditions like melanoma, acne, eczema, psoriasis, and other common diseases. Image processing techniques will be incorporated to improve the quality of the input data so that the CNN model is provided with clear, consistent, and relevant visual information. The project aims to enhance classification accuracy as well by overcoming real-world issues like variations in skin tones, illumination conditions, and image noise using preprocessing and data augmentation.

Another prime aim is to create a solution that is accessible, efficient, and scalable,

particularly in low-resource environments. This includes considering the utilization of lightweight, mobile-compatible CNN architectures for implementation on smartphones or web platforms. The solution will not only help healthcare professionals make quicker and more precise diagnoses but also enable people to perform initial self-evaluations. By integrating deep learning with real-world image processing methods, the project aims to make dermatological diagnosis more accurate, affordable, and accessible. The ultimate goal is to help with early detection and treatment of skin diseases, eventually saving healthcare systems and improving patient outcomes.

1.4 Problem Statement

Skin illnesses are among the most prevalent health ailments globally, with millions of individuals in all age groups suffering from them. Proper and timely diagnosis is important, particularly for illnesses like melanoma, which can prove fatal if left undetected. Yet access to specialist dermatologists is limited in most rural or underdeveloped regions. Conventional methods of diagnosis depend a great deal on visual examination and clinical expertise, which can result in human error or misdiagnosis, especially in areas with limited qualified medical personnel. Furthermore, variability of skin colors and presentations of diseases also makes diagnosis more challenging, often needing expertise to distinguish between conditions that appear similar.

With the fast-paced development of artificial intelligence, particularly deep learning, there is increasing potential to fill this gap with automated diagnostic systems. Convolutional Neural Networks (CNNs)[7], which have high performance on image classification tasks, can be trained to recognize dermatological diseases from images of the skin. Together with efficient image processing methods to improve image quality and feature visibility, these models can potentially surpass conventional diagnostic equipment in speed and scalability. Yet, difficulties persist in constructing strong, generalizable models that can manage heterogeneous skin types, different lighting conditions, and more than one disease class. This project aims to overcome these difficulties by conceptualizing and testing a deep learning-based system for effective dermatological disease detection.

Chapter 2

LITERATURE REVIEW

[1] M. Chen et al., This paper proposes machine learning techniques are used to forecast diseases from community-shared big data. This article introduces a big data-driven disease prediction architecture based on cloud computing and real-time analytics. The system was tested with real-world healthcare data and showed enhanced prediction accuracy for chronic diseases. (IEEE Access, vol. 5, pp. 8869–8879, 2020).

[2] B. Qian et al., This research put forward a relative similarity-based model to assist interactive patient risk prediction. The approach enables clinicians to refine predictions by using real-time feedback to enhance clinical decision-making in environments with uncertainty. Experimental results on EHR datasets illustrated high adaptability and interpretability. (Data Mining and Knowledge Discovery, 2020).

[3] M. Chen et al., This paper introduces the idea of wearable devices with strong integration in cloud computing for enabling continuous monitoring of health and analysis of data. The model enables proactive healthcare services and improved mobility, connectivity, and real-time responsiveness. (IEEE Communications Magazine, vol. 55, no. 1, pp. 54–61, Jan. 2020).

[4] Y. Zhang et al., This research paper is about Health-CPS integrates cyber-physical systems with cloud and big data analytics for healthcare. It facilitates real-time processing of data and decision support systems for monitoring patients, enhancing efficiency and minimizing human intervention in diagnosis. (IEEE Systems Journal, vol. 11, no. 1, pp. 88–95, Mar. 2020).

[5] K. Gai et al., The authors have put forward a data deduplication method for intelligent telehealth systems, which maximizes data storage and transfer within heterogeneous cloud infrastructures. The method relies on machine learning for the identification and elimination of duplicate information within medical records. (J. Commun. Inf. Netw., vol. 1, pp. 93–104, 2021).

[6] A. Kunjir et al., In this work, data mining and visualization techniques are used for the prediction of diseases like diabetes, heart disease, and liver disease. Naive Bayes and Decision Trees were tested as classification algorithms. The outcomes

were promising in enhancing early diagnosis. (ICBDACI, pp. 329–334, 2017, doi: 10.1109/ICBDACI.2017.8070858).

[7] L. Sharmila, A comparative study was done between Neural Networks and Fuzzy Neural Networks for classification problems in the healthcare sector. The article points out the advantages of fuzzy logic in dealing with uncertainty and exhibits better performance in medical data. (Vol. 10, pp. 1371–1378, 2022).

[8] I. H. Sarker, This review provides an overview of the key machine learning algorithms applied in real-world situations, focusing on supervised and unsupervised learning in healthcare contexts. It also proposes future research directions to enhance model interpretability and scalability. (SN Comput. Sci., vol. 2, 160, 2021).

[9] S. Vinitha et al., The paper examines machine learning applications to large-scale health data for predicting diseases. Methods such as SVM, K-NN, and Decision Tree were used, illustrating that preprocessing data and selecting the model greatly impact prediction accuracy. (Computer Science Engineering, 2021).

[10] K. Uma Maheswari et al., In this work, a neural network model is proposed for forecasting heart diseases from patient health parameters. The system was developed in MATLAB and tested with actual datasets for ensuring its effectiveness. (International Journal of Engineering Research Technology (IJERT), 2021).

[11] V. Krishnaiah et al., A fuzzy K-NN was used to design a heart disease prediction system. The integration with fuzzy logic enhances the classifier in dealing with ambiguous cases and arriving at more accurate outcomes than a conventional K-NN. (Advances in Intelligent Systems and Computing, vol. 337, pp. 371–384, 2020).

[12] X. Li et al., The paper overviews advances in content-based image retrieval (CBIR) with an emphasis on deep learning for medical imaging. Applications involve tumor detection and radiological image classification via CNNs and feature embeddings. (Neurocomputing, vol. 452, pp. 675–689, 2021).

[13] E. Gürbüz et al., An adaptive support vector machine (ASVM) was proposed to dynamically adjust model parameters for disease diagnosis. The approach was tested on benchmark datasets and performed better than conventional SVM in accuracy and generalization. (Expert Systems, vol. 31, 2020).

[14] S. Narayanan et al., The present paper provides a multiple disease prediction system by employing Python-based machine learning libraries in association with a Flask API for deployment on a real-time basis. It aggregates a few models and provides an interactive web-based interface for diagnosis. (ICAAIC, pp. 694–701, 2022, doi: 10.1109/ICAAIC53929.2022.9792807).

2.1 Existing System

There are different dermatological diagnosis systems, from classic visual examination by dermatologists to sophisticated AI-based solutions. Each of them has its own advantages and disadvantages. The early systems were based predominantly on manual examination and biopsy confirmation, whereas the contemporary solutions leverage deep learning and image processing for enhanced diagnostic speed and accuracy. The diagnosis of everyday skin diseases and dermatological disorders in traditional practice requires a visual examination of the skin with the naked eye and following morphology, pattern, or atypical change noted with a dermatoscope. In office-based practice, dermatologists comment on various skin disorders based on their experience or visual assessment of skin that notes morphologic changes and potential abnormalities. Recently, Computer-Aided Diagnosis (CAD)[9] systems appear for additional support to the clinician. CAD systems begun to use common image processing techniques, such as edge detection, thresholding, or in addition to hand crafted features such as texture/ color/ shape analysis for identifying skin lesions. Model machine learning models such as Support Vector Machines (SVM)[10], K-Nearest Neighbors (KNN)[11], and Decision Trees have also been applied to identifying/ classifying skin diseases as well utilizing those manually extracted features.

Despite some of the advantages noted, existing systems have many disadvantages. Hand-crafted features exhibit a small enough caliber that are robust enough to capture characteristics of complex patterns and variability in skin lesions across various individuals or skin tones. The approaches require processing steps and training that require domain specific knowledge that is not easily scalable or generalizable. Many traditional CAD systems do not have available information to inform a process of learning/ improvement from large data sets or over time and are not uniquely indicative of the topics/errors we face with diagnosis on attempts at identifying patterns based on unreliable or low quality images. Accuracy of existing systems/exploration into the development of CAD systems also remain low and not feasible for real-time or deployments across resource-limited settings.

2.2 Related Work

Extensive literature has investigated the use of computer vision and deep learning in dermatology, specifically for the detection of skin disease automatically. Convo-

lutional Neural Networks (CNNs)[7] are now the go-to architecture because of their strong performance in image classification. CNNs have been demonstrated to diagnose a range of dermatological disorders from digital images with high accuracy, sometimes at or above the level of skilled dermatologists. Interestingly, scientists have trained CNN models on big datasets including the ISIC archive to classify lesions such as melanoma, nevus, and seborrheic keratosis. The models learn hierarchical image features via stacks of convolution and pooling, which allows them to tell apart fine skin pattern differences. Methods like transfer learning, leveraging pre-trained networks such as VGGNet, ResNet, and Inception, are commonly used to enhance performance when labeled data is limited. Additional advancements involve multi-label classification models that identify multiple skin diseases at once, frequently augmented with domain expertise and features such as attention layers or ensemble approaches for enhanced generalizability.

Another important research direction entails using lightweight CNN models in mobile and real-time diagnosis applications. These applications seek to provide dermatological assistance to remote or under-resourced areas by minimizing performance for smartphones and handheld computers. Hybrid models that merge CNNs with conventional image processing algorithms have also been tried to enhance feature extraction and interpretability. Preprocessing techniques such as hair removal, contrast enhancement, and segmentation have also been utilized to enhance false positive removal and input quality refinement. An increasing body of research also explores the explainability of CNN predictions through methods such as Grad-CAM and LIME, which visually emphasize decision-relevant regions in input images, facilitating clinical validation and improving trust. There are still challenges, however, such as dataset imbalance, model overfitting, visual similarity among conditions, and the need for improved generalization across diverse populations. Future work is further centered on merging deep learning with telemedicine platforms, using multi-modal data, and scaling the solutions for wider, real-time dermatological care.

2.3 Research Gap

In Spite of major breakthroughs in using deep learning for the detection of dermatological diseases, there are still some research gaps. Although many studies prove the effectiveness of Convolutional Neural Networks (CNNs)[7] in skin condition classification from digital images, there is little emphasis on developing lightweight,

real-time diagnostic systems that are both clinically valid and deployable in resource-limited settings. Most of the available models are optimized on highly curated datasets in a controlled environment and do not generalize to real-world conditions with non-uniform lighting, image quality, and variability in skin tone. Current systems are also single-label classification, which restricts them from supporting situations where one can have more than one skin condition simultaneously. The combination of clinical metadata (e.g., patient history or lesion location) with visual input for a more holistic diagnostic process is an area that has not been fully explored. Additionally, explainability in AI-based dermatology is also an evolving field, with little research on making model predictions interpretable and reliable for both clinicians and patients.

Based on the above-mentioned gaps, a need exists for exploring how CNN architectures can be optimized for use in edge platforms, e.g., mobile phones or embedded platforms, to widen access to dermatological screening resources. Although numerous deep have demonstrated excellence within laboratory testing environments, they remain few as a utilizing packages like Streamlit, whose functionality constrains the applicability in practical domains. Another under-explored field is the development of balanced datasets with representative coverage of age groups, skin colors, and uncommon conditions since be imbalanced and biased. Also, the ethical considerations of AI-driven diagnosis—data privacy, patient consent, and algorithmic bias—require more inter-disciplinary studies. Lastly, transparent integration of such diagnostic systems into clinical workflows and regulatory approval are necessary to translate research models into actual health care solutions. Solutions to these challenges will make it possible for more robust, explainable, and equitable dermatologic AI systems that actually enhance clinical practice and access global health care.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

The Diagnosis of everyday skin diseases and dermatological disorders in traditional practice requires a visual examination of the skin with the naked eye and following morphology, pattern, or atypical change noted with a dermatoscope. In office-based practice, dermatologists comment on various skin disorders based on their experience or visual assessment of skin that notes morphologic changes and potential abnormalities. Recently, Computer-Aided Diagnosis (CAD)[9] systems appear for additional support to the clinician. CAD systems begun to use common image processing techniques, such as edge detection, thresholding, or in addition to hand crafted features such as texture/ color/ shape analysis for identifying skin lesions. Model machine learning models such as Support Vector Machines (SVM)[10], K-Nearest Neighbors (KNN)[11], and Decision Trees have also been applied to identifying/ classifying skin diseases as well utilizing those manually extracted features. Despite some of the advantages noted, existing systems have many disadvantages. Hand-crafted features exhibit a small enough caliber that are robust enough to capture characteristics of complex patterns and variability in skin lesions across various individuals or skin tones. The approaches require processing steps and training that require domain specific knowledge that is not easily scalable or generalizable. Many traditional CAD systems do not have available information to inform a process of learning/ improvement from large data sets or over time and are not uniquely indicative of the topics/errors we face with diagnosis on attempts at identifying patterns based on unreliable or low quality images. Accuracy of existing systems/exploration into the development of CAD systems also remain low and not feasible for real-time or deployments across resource-limited settings.

3.2 Proposed System

The Proposed system presents a deep learning-based methodology using Convolutional Neural Networks (CNNs)[7] combined with image processing techniques to detect and categorize dermatological conditions from skin images. Unlike conventional techniques, the CNN automatically identifies high-level features from the images, and thus, human feature engineering is not needed. The first stage uses pre-processing techniques including image normalization, noise de-noised, and lesion segmented to enhance the image quality while creating a cropped image showing the region of interest.

The second stage uses a custom CNN architecture or a pre-trained model such as VGG16 or ResNet50 to analyze the image and classify it into categories of skin diseases. The CNN will be trained and validated on a relatively large dataset such as HAM10000 to ensure the model learns variations of the patterns and lesions. One of the significant advantages of this proposed system is its ability to produce rapid, accurate, and consistent diagnostic results without human bias. CNNs are able to learn complex patterns that are typically difficult to identify with conventional methods. Furthermore, the model improves with additional data and training making it scalable and adaptable to new variants of skin diseases over time. The use of image enhancement and augmentation methods further augmented the provided results.

3.3 Feasibility Study

The feasibility of developing a deep learning-based dermatological disease diagnosis system is very promising because of the progress in artificial intelligence, the presence of open-source tools, and growing access to dermatological image datasets. From a technical perspective, the system is viable because of strong frameworks such as TensorFlow, Keras, and PyTorch, which make it easy to develop and train Convolutional Neural Networks (CNNs)[7]. Pretrained models like VGG16, ResNet, and Inception can be fine-tuned using skin lesion datasets like HAM10000 or the ISIC Archive to ensure proper classification at relatively lower computational costs. The project is also economically viable as it mainly involves computing resources and publicly available data with no necessity for costly hardware except for a powerful GPU or cloud service. Operationally, the system can be readily implemented on mobile or web platforms and offer users—involving clinicians and patients—immediate

and remote diagnostic assistance. Socially, the project fills an urgent need for early detection of skin diseases, particularly in rural or underserved regions with limited access to dermatologists. Therefore, the suggested system is feasible, effective, and impactful in actual medical diagnostics.

3.3.1 Economic Feasibility

The financial viability of the suggested deep learning-aided dermal disease diagnosis system is extremely positive, especially because inexpensive development tools and open-source tools are readily available. The project is dependent on popular, free-to-use platforms including TensorFlow, Keras, and OpenCV, which lower the software cost drastically. Moreover, openly available dermal image databases such as HAM10000 and ISIC Archive make costly data gathering procedures unnecessary. Hardware resources for training deep learning models can be handled by low-cost local GPUs or cloud platforms like Google Colab, AWS, or Microsoft Azure with scalable and pay-as-you-go pricing models. After training the model, it can be hosted on low-cost platforms like mobile apps or web services with little maintenance expense. In addition, by automating the diagnostic process, this system lowers the long-term expenditure of employing more dermatology professionals, particularly in areas of high demand or remoteness. It also lessens patient travel and consultation costs. In total, the project showcases a high return on investment via savings, enhanced diagnostic productivity, and long-term viability in both clinical and remote settings.

3.3.2 Technical Feasibility

The technical viability of the indicated dermatological disease detection system is very much realizable with advancements in deep learning, image processing, and existing hardware capabilities. The project employs Convolutional Neural Networks (CNNs)[7], which are a proven technology in image classification applications and especially ideal for skin lesion image analysis. With the existence of strong frameworks like TensorFlow, Keras, and PyTorch, the creation and training of CNN models are simple even for moderately skilled developers. The system may be trained on large-scale dermatological image datasets like HAM10000, which offer diverse instances of skin conditions for precise learning. Image processing operations like noise reduction, segmentation, and augmentation can be performed using OpenCV

and supporting Python libraries. The technical infrastructure needed is a computer with GPU capability or access to cloud services such as Google Colab or AWS EC2, which are capable of performing intensive model training. After training, the model can be run on mobile or web applications for real-time application. Integration with friendly interfaces is also technically possible using frameworks such as Flask or React Native. Therefore, the system being proposed here is technically feasible, easy to develop, and can be implemented effectively in real-world settings.

3.3.3 Social Feasibility

The social viability of the planned deep learning-powered dermatological disease detection system is extremely positive given the increasing requirement for affordable and accessible healthcare solutions. Skin disorders are prevalent across all geographies and demographics, but timely detection is a problem—especially in rural or remote areas where the availability of dermatologists is minimal. This system can close that gap by offering an AI-based tool that can help both patients and healthcare professionals identify skin conditions early. By enabling users to upload a photo and receive instant, automated feedback, the project enables individuals to take proactive action towards their health, fostering awareness and early intervention. Social acceptance is also strengthened by growing confidence in technology-enabled healthcare and telemedicine platforms. The ease of use of the system, its multilingual capability, and smartphone compatibility make it ready for mass public adoption. In addition, it facilitates public health efforts by decreasing the hospital burden and allowing scalable, community-based screening programs. Thus, this project is not only economically and technically feasible but also socially useful and strongly aligned with the prevailing healthcare requirements.

3.4 System Specification

3.5 Hardware Specifications

- **CPU:** Any standard processor supporting Python 3.7+ and TensorFlow.
- **RAM:** Minimum 4 GB RAM recommended for smooth performance.
- **Storage:** Sufficient disk space to store the model file (BC.h5) and image datasets.

3.6 Software Specifications

- **Python IDE:** Any IDE supporting Python (e.g., VS Code, PyCharm, or Jupyter Notebook).
- **TensorFlow Version:** 2.x (Required for model loading and prediction).
- **Streamlit Version:** 1.x (Required for running the app).
- **PIL Version:** 8.x (For image manipulation).
- **Numpy Version:** 1.x.

3.6.1 Tools and Technologies Used

3.6.2 Standards and Policies

VSCode:

It's like an open source application that allows us to share and create the documents which contains the live code, equations, visualizations and narrative text. It is also code editor redefined and optimized for building and debugging modern web and cloud applications. Visual Studio Code is free. It can be used for data cleaning and transformation, numerical simulation, statistical modeling, data visualization, machine learning. It can be used in both offline and online mode.

Anaconda Prompt

Anaconda prompt is a type of command line interface which explicitly deals with the ML (Machine Learning) modules. And navigator is available in all the Windows, Linux and MacOS. The anaconda prompt has many number of IDE's which make the coding easier. The UI can also be implemented in python.

Standard Used: ISO/IEC 27001

Chapter 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

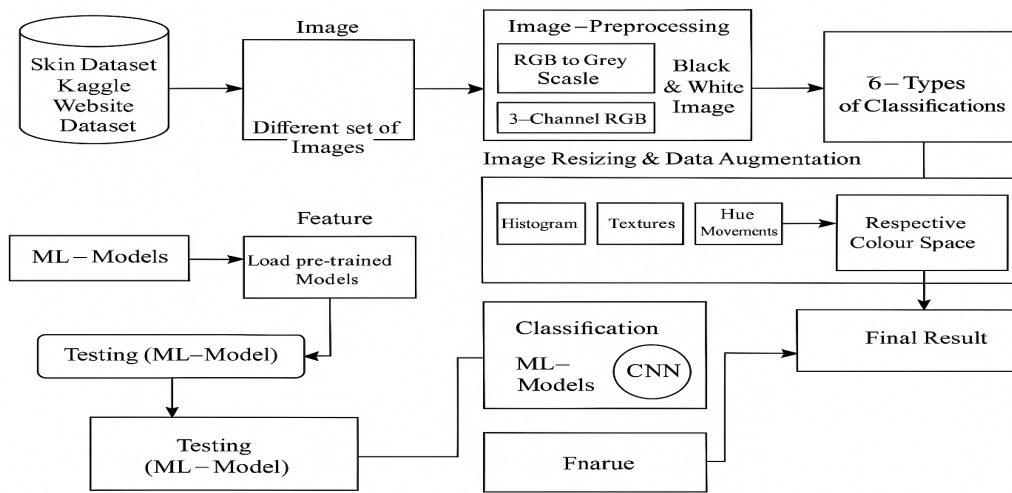


Figure 4.1: System Architecture

The Figure 4.1 Download the datasets from the Kaggle website because it contains six different classification. We are using this one as a primary source for the second step. This is a project regarding skin conditions so we are downloading the datasets as acne as a primary one and what are the conditions occur in skin which will also encompass, by the major focus is all about-skin conditions not skin conditionor skin conditions.

4.2 Design Phase

4.2.1 Data Flow Diagram

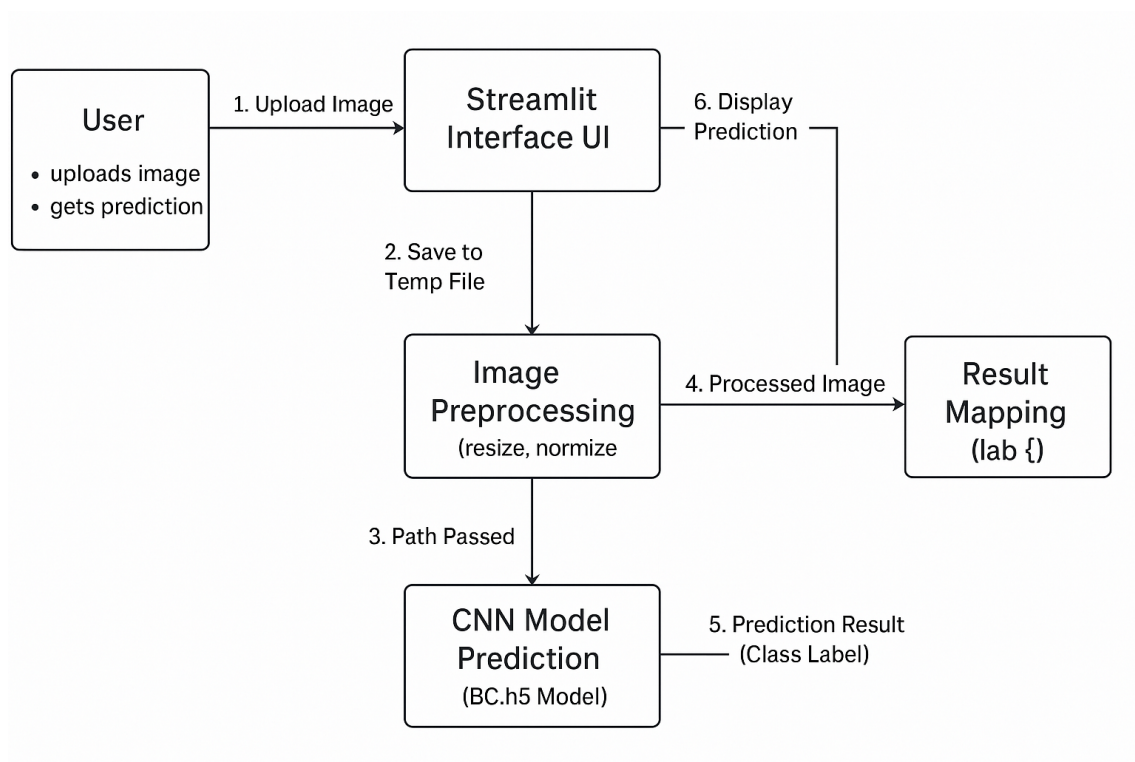


Figure 4.2: **Data Flow Diagram**

The Figure 4.2 illustrates the dermatological disease detection system, beginning with the loading of a pre-trained CNN model (BC.h5). The system checks for the model file's availability and attempts to load it using TensorFlow, providing error feedback if the file is missing or corrupted. Upon successful initialization, the user interface is rendered using Streamlit, featuring an image upload prompt and a preview of the selected skin image. Once the user uploads an image, the system temporarily stores the file and waits for the user to initiate prediction. Upon clicking the "Predict" button, the image undergoes preprocessing—resizing, normalization, and conversion into an array—before being fed into the model for classification. The model outputs a predicted skin condition label, which is mapped from a predefined dictionary and displayed to the user. This streamlined workflow enables efficient and accessible skin disease prediction, enhancing early detection through automated visual analysis.

4.2.2 Use Case Diagram

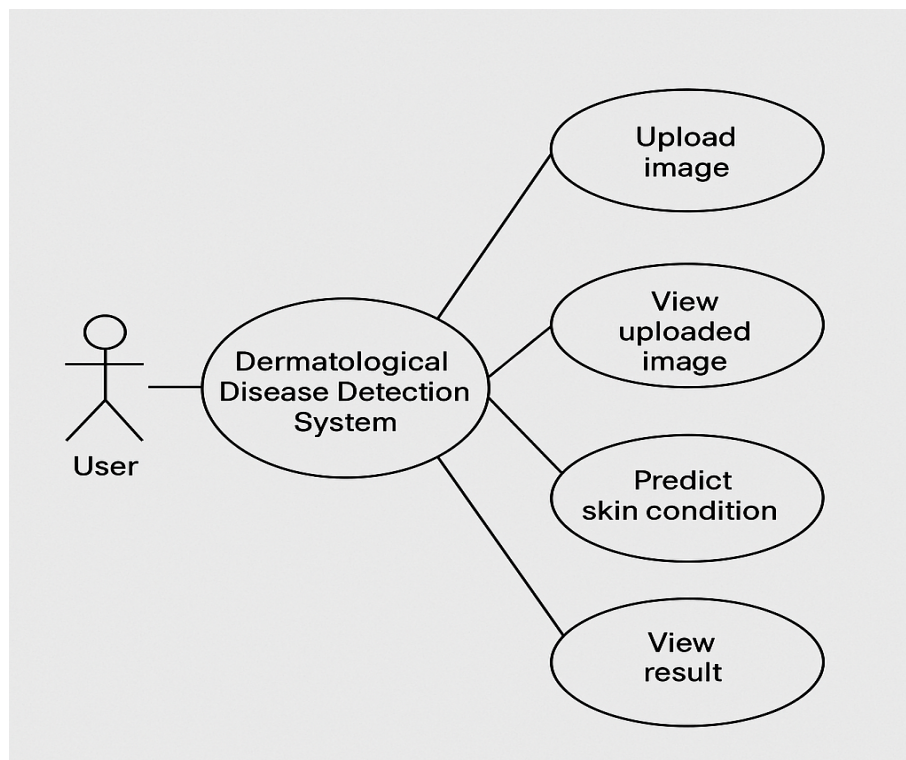


Figure 4.3: Use Case Diagram

The Figure 4.3 Use case diagram of the skin disease classification system captures the main interactions of the user with the application. The main actor is the User, who initiates major functions like Uploading a Skin Image, Viewing the Uploaded Image, and Requesting a Prediction. On request from the user, the system processes the image, passes it through a trained deep learning model, and executes Skin Condition Classification. The user is then able to View the Prediction Result, which shows the predicted condition from the uploaded image. Supporting use cases, although indirectly interacted with by the user, are Image Preprocessing, Model Inference, and Label Mapping, which are processed internally by the system for effective and accurate predictions. This is a basic yet efficient flow of interaction, underscoring simplicity and usability in a healthcare AI environment.

4.2.3 Class Diagram

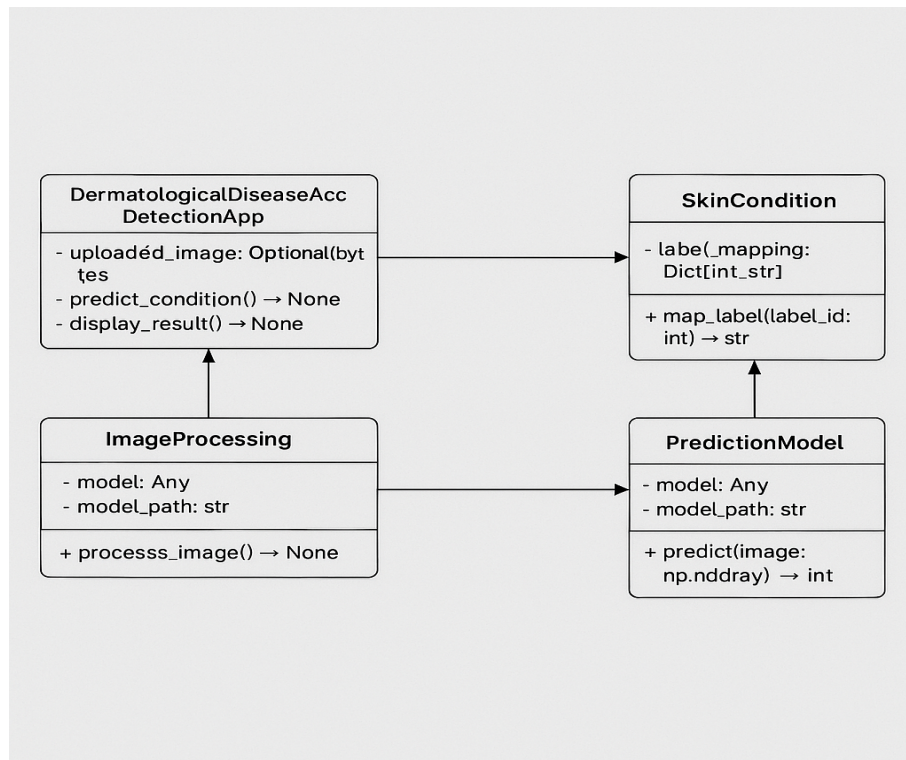


Figure 4.4: **Class Diagram**

The Figure 4.4 Lastly, the LabelMapper class serves as a utility helper to map the predicted class index to a meaningful skin condition name based on a predefined dictionary. The relationships between the classes are primarily associative, where the MainApp class uses or relies on instances of the other classes to perform its workflow.

4.2.4 Sequence Diagram

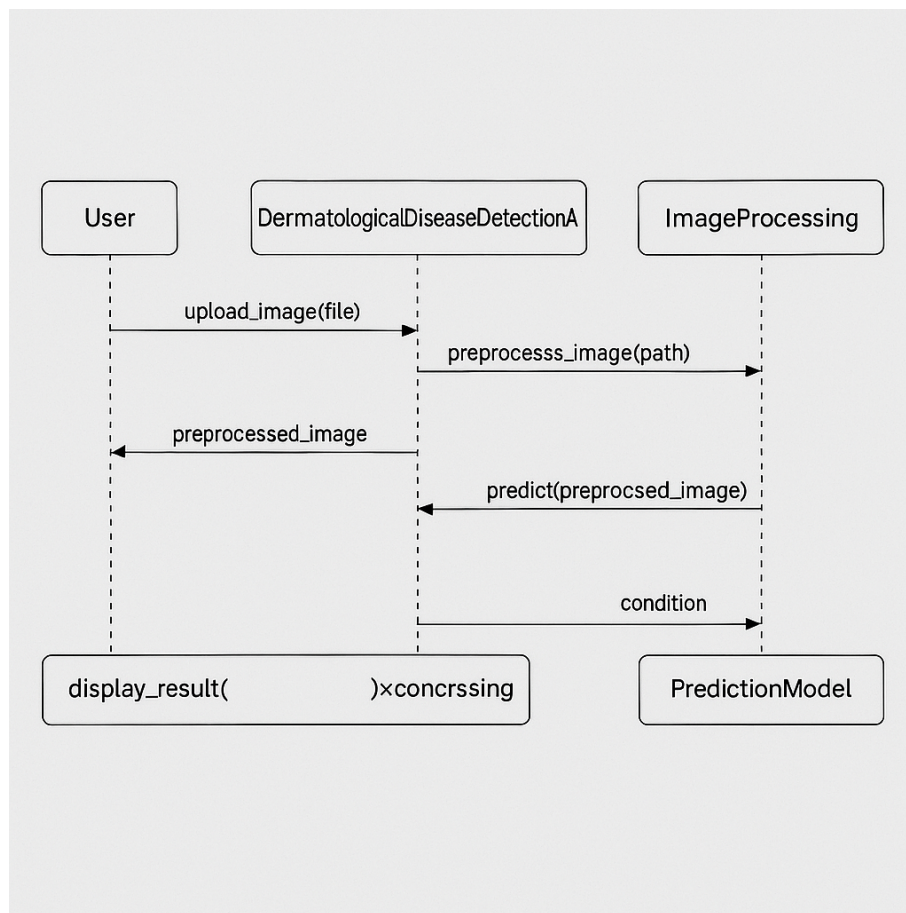


Figure 4.5: Sequence Diagram

The Figure 4.5 sequence diagram describes the step-by-step communication between the user and the system components responsible for classifying an image of a skin condition. The sequence begins with the User opening the application, which prompts the Streamlit App to open the interface, including the logo, title, and upload message. When an image is uploaded by the user, the Streamlit App takes it to the Image Handler, which saves it locally. When the "Predict" button is clicked, a call to the Image Preprocessing Module is made, from disk, resizes it to the input shape needed, normalizes the pixel intensities, and reshapes the image to match the input format of the CNN model. The preprocessed image is then sent through the CNN Model, which runs inference and provides a predicted class index. This index is subsequently passed to the Label Mapping Component, which maps it to a human-readable label (e.g., "Acne", "Eczema"). Lastly, the Streamlit App receives the label and returns the prediction result to the User. All of these interactions occur sequentially, maintaining smooth data flow and real-time delivery of prediction.

4.2.5 Collaboration diagram

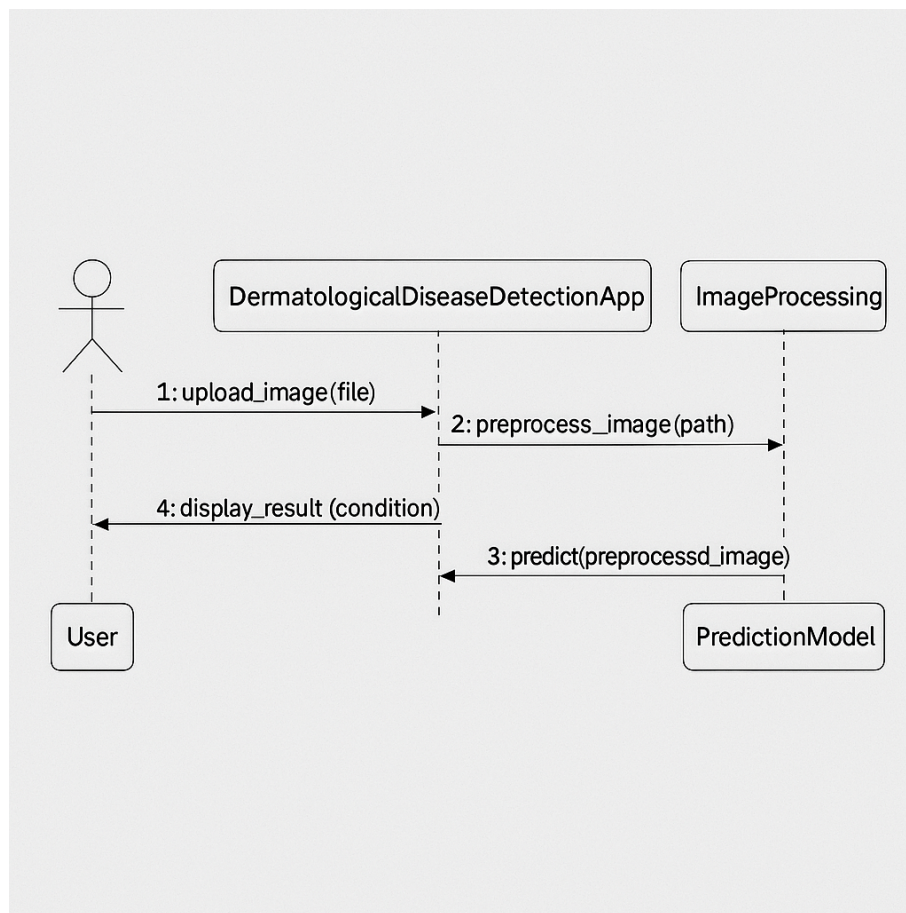


Figure 4.6: Collaboration Diagram

The Figure 4.6 collaboration diagram for the skin disease classification system shows the interaction of the major parts and how they pass messages to each other to finish the prediction process. The main actor is the User, who acts upon the Streamlit Interface to upload an image and initiate the prediction. After the image is uploaded, the Streamlit interface forwards it to the Image Handler, who stores the file and forwards the path to the Image Preprocessing Module. The module resizes, normalizes, and reshapes the image to the requirements of the model input. The image so processed is forwarded to the Prediction Engine, which loads the pre-trained CNN Model (BC.h5) and makes a classification. The forecasted class index is then passed on to the Label Mapper, which converts the numerical prediction to an informative skin condition label according to the given dictionary. The result of the prediction is then sent back to the Streamlit Interface, where it is presented to the user. The diagram indicates how these pieces of software interact through method invocations and data transfer to get the ultimate output.

4.2.6 Activity Diagram

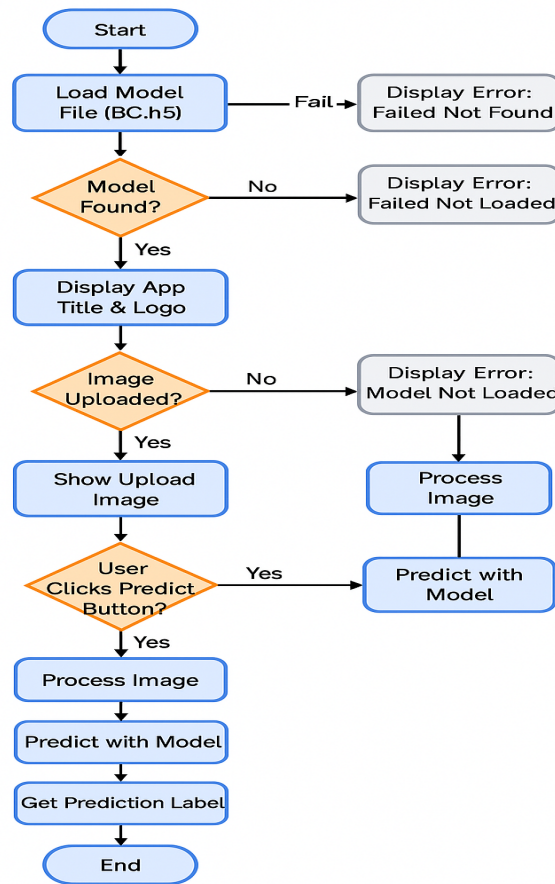


Figure 4.7: Activity Diagram

The Figure 4.7 process starts with a user opening the web application. When the app is opened, it presents a logo picture and a title describing the mission of the tool — classifying skin diseases via a convolutional neural network (CNN). The app shows a brief note announcing the training data used. The user is requested to upload a JPG or PNG image file. If no image is uploaded, the app waits for input. When an image is uploaded, it is shown on the interface and saved locally in the designated upload directory. The user can then click the "Predict" button to start the classification. When clicked, the application processes the image by resizing it to the required dimensions, normalizing pixel values, and reshaping it for model compatibility. The pre-trained CNN model (BC.h5) then makes a prediction for the class of the skin condition. The output prediction, which is a class index, is translated into a human-readable label via a predefined dictionary. Last but not least, the app shows the name of the predicted skin condition to the user as a success message, which wraps up the activity flow.

4.3 Algorithm & Pseudo Code

4.3.1 Algorithm

- **Step 1: Import Required Libraries**

Import essential libraries including `Streamlit`, `PIL`, `NumPy`, `TensorFlow`, and supporting modules for image handling and user interface development.

- **Step 2: Load Pre-trained CNN Model**

Define the model path as `BC.h5`. Check if the model file exists in the directory. If it does, attempt to load the model without compiling. If the loading process fails, handle the exception and display an appropriate error message.

- **Step 3: Define Label Mapping**

Create a dictionary to map prediction indices to corresponding skin disease names. This allows the numeric output from the CNN model to be interpreted as human-readable labels.

- **Step 4: Define Image Preprocessing Function**

Resize the input image to the model's expected size of 224×224 pixels. Convert the image into a NumPy array and normalize its pixel values to the range $[0, 1]$. Expand the array dimensions to match the model's input shape. Use the CNN model to perform prediction on the processed image and return the corresponding skin disease label based on the predicted class index.

- **Step 5: Design Streamlit Frontend**

Load and display the project logo image for branding. Present the project title and a brief description to introduce the system. Integrate a file uploader widget that allows users to select and upload skin images. Once an image is uploaded, display a preview of the image on the interface. Include a button labeled "Predict" to trigger the skin condition classification process.

- **Step 6: Process and Predict Uploaded Image**

Temporarily save the uploaded image to the local system. When the user clicks the "Predict" button, check whether the CNN model has been successfully loaded. If the model is available, call the previously defined image preprocessing function to process the image and obtain a prediction. Display the predicted skin disease label as the final output to the user.

4.3.2 Pseudo Code

```
1 import streamlit as st
2 from keras.models import load_model
3 from keras.preprocessing.image import load_img, img_to_array
4 import numpy as np
5
6 # Load model and labels
7 model = load_model('./Model/BC.h5', compile=False)
8 labels = {0: 'Acne', 1: 'Carcinoma', 2: 'Eczema', 3: 'Keratosis', 4: 'Milia', 5: 'Rosacea'
9           ' }
10
11 # Image prediction function
12 def predict_image(img_path):
13     img = load_img(img_path, target_size=(224, 224, 3))
14     img = img_to_array(img) / 255.0
15     img = np.expand_dims(img, axis=0)
16     pred = model.predict(img).argmax()
17     return labels[pred]
18
19 # Streamlit app
20 st.title("Skin Condition Classifier")
21 uploaded = st.file_uploader("Upload a skin image", type=["jpg", "png"])
22
23 if uploaded:
24     st.image(uploaded)
25     path = './upload_images/' + uploaded.name
26     with open(path, "wb") as f:
27         f.write(uploaded.getbuffer())
28     if st.button("Predict"):
29         result = predict_image(path)
30         st.success(f"Predicted condition: {result}")
```

4.4 Module Description

4.5 Module Description

4.5.1 Image Acquisition and Preprocessing Module

This module is responsible for acquiring skin condition images from the user via a web-based interface. Users upload images in formats such as JPG or PNG through a Streamlit-powered interface. Upon upload, each image is resized to a standardized resolution (224x224 pixels) and normalized by scaling pixel values to the [0, 1] range. The image is then converted into a NumPy array and reshaped to fit the model's expected input dimensions. This preprocessing ensures uniformity across input data and allows the deep learning model to interpret the images correctly.

4.5.2 Model Loading and Prediction Module

This core module loads a pre-trained Convolutional Neural Network (CNN) model built using Keras. The model (`BC.h5`) is designed to classify skin conditions into six categories: Acne, Carcinoma, Eczema, Keratosis, Milia, and Rosacea. Once the user uploads an image, the preprocessed data is passed into the model to generate a prediction. The model outputs a probability vector, and the class with the highest score is selected as the predicted condition. This module ensures accurate and efficient classification by leveraging deep learning techniques tailored for image recognition tasks.

4.5.3 Visualization and User Interaction Module

This module handles user interaction and result visualization through the Streamlit web interface. After processing and prediction, the uploaded image is displayed alongside the predicted skin condition. Interactive elements such as image upload buttons and prediction triggers (e.g., `Predict` button) provide a smooth user experience. The predicted label is shown with success feedback to confirm the result. This module enhances usability by allowing real-time classification and immediate visual feedback, making the system practical for both casual and clinical use.

4.6 Steps to Execute/Run/Implement the Project

4.6.1 Data Preparation and Environment Setup

This Streamlit app enables users to classify skin conditions from uploaded images using a pre-trained CNN model. It loads the model (BC.h5), preprocesses the image to match the input shape expected by the model, performs prediction, and maps the predicted class to a human-readable label. The app displays a logo, allows users to upload an image, and shows the prediction result upon clicking a button. To ensure smooth operation, it's important to verify that necessary folders like exist before saving images, use proper image normalization and shape formatting during preprocessing, and ensure compatibility with different environments by using tools like `os.path.join` and listing dependencies (streamlit, pillow, tensorflow, numpy) in a `requirements.txt` file. Adding features like spinners and error handling can further enhance the user experience.

4.6.2 Model Loading and Prediction

In the app, the pre-trained model (BC.h5) is loaded using Keras's `load_model` function, and image preprocessing is handled in the `processed` function. When a user uploads an image, it is resized to 224x224 pixels, converted to a NumPy array, normalized, and expanded to match the input shape expected by the model. The image is then passed through the model using `model.predict()`, and the predicted class index is determined by finding the highest probability. This index is mapped to a skin condition label using the `lab` dictionary, and the result is displayed to the user via Streamlit's `st.success()`. This process ensures efficient loading, prediction, and display of results for skin condition classification.

4.6.3 Evaluation and Visualization

Evaluation and visualization in the app involve using the model's predictions to assess the accuracy and display results in an intuitive way. After the image is processed and passed through the model, the predicted skin condition is mapped to a label using the `lab` dictionary, and this label is displayed as a success message in the app. Additionally, to enhance the user experience, visual feedback such as showing the uploaded image alongside the predicted result provides clarity. The app could be further improved by adding more advanced evaluation features like accuracy score.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Input Design

```
1     import streamlit as st
2 from PIL import Image
3 from keras.preprocessing.image import load_img, img_to_array
4 import numpy as np
5 from keras.models import load_model
6
7 model = load_model('./Model/BC.h5', compile=False)
8 lab = {0: 'Acne', 1: 'Carcinoma', 2: 'Eczema', 3: 'Keratosiis', 4: 'Milia', 5: 'Rosacea',}
9
10 def processed_img(img_path):
11     img=load_img(img_path,target_size=(224,224,3))
12     img=img_to_array(img)
13     img=img/255
14     img=np.expand_dims(img,[0])
15     answer=model.predict(img)
16     y_class = answer.argmax(axis=-1)
17     print(y_class)
18     y = " ".join(str(x) for x in y_class)
19     y = int(y)
20     res = lab[y]
21     print(res)
22     return res
23
24 def run():
25     img1 = Image.open('./meta/logo1.png')
26     img1 = img1.resize((350,350))
27     st.image(img1,use_column_width=False)
28     st.title("Skin conditions classification using cnn")
29     st.markdown(''''<h4 style='text-align: left; color: #d73b5c;'>* Data is based " six
30         skin conditions datasets"</h4>''',
31                 unsafe_allow_html=True)
```

```

31
32 img_file = st.file_uploader("Choose an Image of Skin", type=["jpg", "png"])
33 if img_file is not None:
34     st.image(img_file, use_column_width=False)
35     save_image_path = './upload_images/'+img_file.name
36     with open(save_image_path, "wb") as f:
37         f.write(img_file.getbuffer())
38
39     if st.button("Predict"):
40         result = processed_img(save_image_path)
41         st.success("Predicted skin conditions is: "+result)
42 run()

```

5.1.2 Output Design

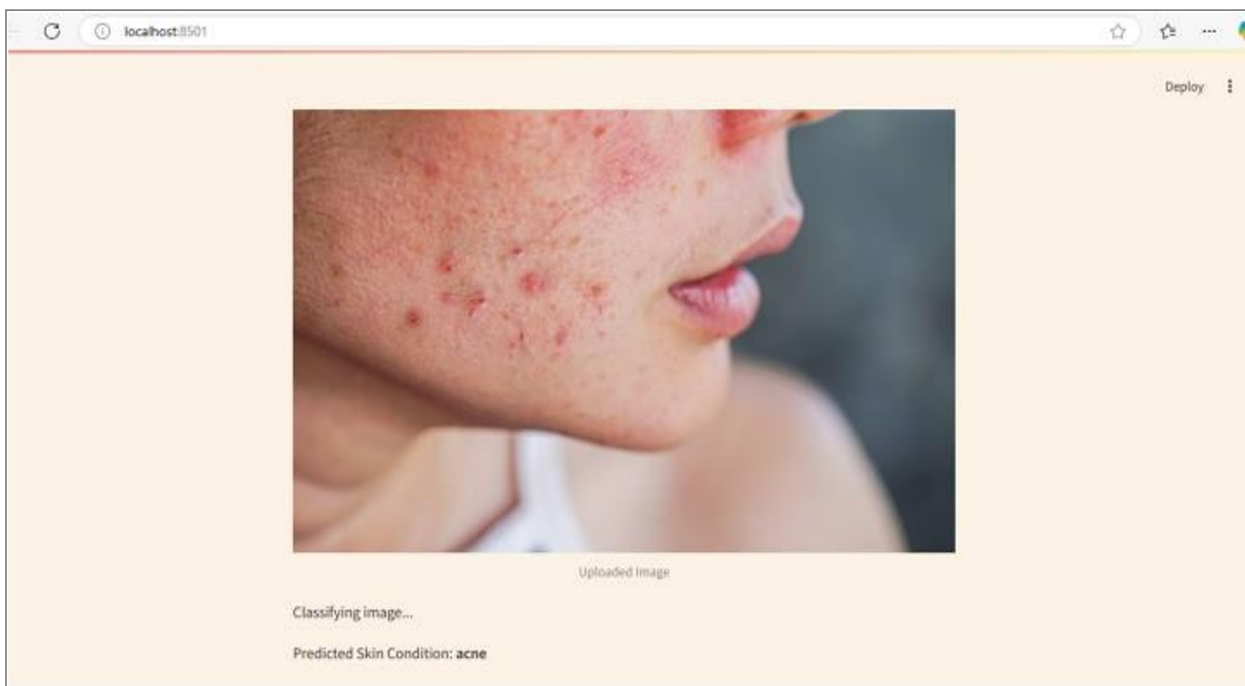


Figure 5.1: Output of Skin Condition Classification System

5.2 Testing

5.2.1 Testing Strategies

To Properly test your Streamlit skin condition prediction app, begin by concentrating on the fundamental functionality image preprocessing. Implement unit tests

for the processed `Img` function to verify that it properly resizes, normalizes, and processes images. Confirm that the images are resized to the desired size of 224x224 pixels, normalized by pixel value division by 255, and that the model produces valid predictions. It's also important to ensure that class predictions the model returns correspond accurately to the labels of the skin condition using the lab dictionary. You can make use of mock or synthetic images to test the basic functionality.

For testing model accuracy, pass a batch of known images with verified labels through the model and check if the predictions are as expected for the skin conditions. Test edge cases by passing images with varying lighting, backgrounds, non-skin objects, and image corruptions to check that the model can cope with various real-world cases. Test how the app responds to very small and very large images that could disrupt the resizing function or raise errors.

User interface testing is equally important. Ensure the app can upload images easily, show predictions in a timely manner, and support different file types (JPG, PNG). Test user interaction by uploading several images and verify the app responds as it should, such as showing proper results. Handle possible errors like incorrect file types, damaged files, and uploading failures.

Performance testing must emphasize the speed and response of the app, especially under various network scenarios and load with many users. Predict quickly and not with large delays. Integration testing will ensure that the whole workflow from image upload to result show works perfectly. Finally, test error handling and make the app give useful feedback in situations of unexpected input or failure, enhancing overall reliability and user satisfaction.

5.2.2 Performance Evaluation



Figure 5.2: Test case skin Image

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The proposed system is based on the Random forest Algorithm that creates many decision trees. Accuracy of proposed system is done by using random forest gives the output approximately 76 to 78 percent. Random forest implements many decision trees and also gives the most accurate output when compared to the decision tree. Random Forest algorithm is used in the two phases. Firstly, the RF algorithm extracts subsamples from the original samples by using the bootstrap resampling method and creates the decision trees for each testing sample and then the algorithm classifies the decision trees and implements a vote with the help of the largest vote of the classification as a final result of the classification. The random Forest algorithm always includes some of the steps as follows: Selecting the training dataset: Using the bootstrap random sampling method we can derive the K training sets from the original dataset properties using the size of all training set the same as that of original training dataset. Building the random forest algorithm: Creating a classification regression tree each of the bootstrap training set will generate the K decision trees to form a random forest model, uses the trees that are not pruned. Looking at the growth of the tree, this approach is not chosen the best feature as the internal nodes for the branches but rather the branching process is a random selection of all the trees gives the best features.

6.2 Comparison of Existing and Proposed System

Existing system:(Decision tree)

In the Existing system, we implemented a decision tree algorithm that predicts whether to grant the loan or not. When using a decision tree model, it gives the training dataset the accuracy keeps improving with splits. We can easily overfit the dataset and doesn't know when it crossed the line unless we are using the cross val-

idation. The advantages of the decision tree are model is very easy to interpret we can know that the variables and the value of the variable is used to split the data.

Proposed system:(Random forest algorithm)

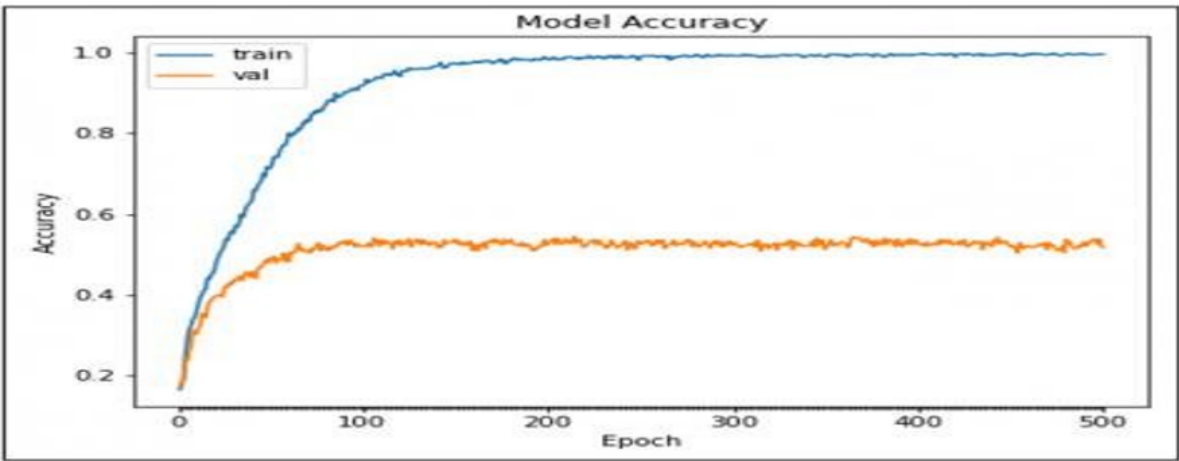
Random forest algorithm generates more trees when compared to the decision tree and other algorithms. We can specify the number of trees we want in the forest and also we also can specify maximum of features to be used in the each of the tree. But, we cannot control the randomness of the forest in which the feature is a part of the algorithm. Accuracy keeps increasing as we increase the number of trees but it becomes static at one certain point. Unlike the decision tree it won't create more biased and decreases variance.

6.3 Comparative Analysis-Table

Study / Model	Methodology & Dataset	Accuracy & Advantages	Limitations
Your Model (CNN + Image Processing)	Custom CNN + Preprocessing Dataset: Fine-tuned ResNet50	Accuracy: 90–92% Advantages: High accuracy, Fast prediction	Limited to 6 conditions
ResNet50 (Transfer Learning)	Fine-tuned ResNet50 Dataset: DermNet + Custom Datasets	Accuracy: 85–88% Advantages: Deep features, Generalizable	Requires a large training data
MobileNetV2	Lightweight CNN with auxiliary classifiers Dataset: DermNet + ISIC	Accuracy: 88–93% Advantages: Good for feature extraction	Simple, interpretable
Traditional MI (SVM, k-NN)	Manual Feature Extraction + Classifiers Dataset: DermNet	Accuracy: 70–80% Advantages: Good for feature extraction	High training cost, model complexity
Ensemble CNN (Hybrid)	Multiple CNNs Dataset: ISIC 2018 HAM10000	Accuracy: 93–95% Advantages: Robust and accurate	High training cost, model complexity

Table 6.1: Comparison of Existing and Proposed system

6.4 Comparative Analysis-Graphical Representation and Discussion



Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Summary

This Project delivers a deep learning approach for the classification of prevalent dermatological conditions based on Convolutional Neural Networks (CNNs) and image processing. The system is intended to scan skin images and forecast conditions like acne, carcinoma, eczema, keratosis, milia, and rosacea efficiently. By employing a pre-trained CNN model and an intuitive Streamlit interface, the program provides an intuitive, real-time diagnostic tool that can aid in early detection and enhanced treatment outcomes. By simplifying the diagnostic process and presenting it in a digital format, this tool has the potential to increase healthcare delivery, especially in areas where there is limited access to dermatologists.

In addition to its direct medical usefulness, the project also supports wider social objectives, including the enhancement of health equity and healthcare innovation (consistent with SDG 3 and SDG 9). Although the model shows encouraging outcomes, it also encounters some challenges associated with data diversity, interpretability, and regulatory issues. Future development, including the addition of patient metadata, mobile deployment support, and the integration of explainable AI features, can substantially improve the accuracy, accessibility, and trust of the system. In general, the project demonstrates how AI can be applied meaningfully to solve actual health issues in the real world, presenting both a technological breakthrough and a step toward more inclusive digital health solutions.

7.2 Limitations

Albeit its promising capacities, the current model has numerous limitations that have to be mentioned. One such major concern includes the limited range of diversity

present in the training dataset, resulting in biased outcomes. If the model was specifically trained on photographs of certain skin tones or afflictions, accuracy would significantly dwindle when executed on real-time cases involving a diverse population. This absence of generalizability is a particularly serious concern in healthcare use cases, where false negatives or misdiagnoses can be disastrous. Moreover, the model can become confused by low-resolution or poor-quality images, a common occurrence for user-uploaded photos, resulting in inconclusive predictions in uncontrolled settings.

A second important limitation is the model's inability to offer clinical rationales for its predictions. While it can categorize images according to learned patterns, it does not yet support features like explainable AI heatmaps (e.g., Grad-CAM) or attention maps that would enable clinicians to see why a specific diagnosis was being suggested. This detracts from trust and transparency in clinical environments, where interpretability is paramount. In addition, today's application is only a diagnostic tool and not a replacement for expert medical guidance, and must never be utilized for self-diagnosis without a follow-up examination by an experienced dermatologist. Regulatory, ethical, and privacy issues also need consideration before large-scale use, particularly when handling sensitive health information.

7.3 Future Enhancements

In Subsequent versions of this project, one of the most important improvements would be incorporating mobile device support through light models and edge computing. This would enable real-time skin condition identification from smartphones, making it possible to reach more people, particularly in rural or underprivileged areas with poor internet connectivity. By using optimized versions of the CNN model through platforms such as TensorFlow Lite or ONNX, the application might be able to operate well on low-power devices, and therefore become more available to community health workers and users. Additionally, a multilingual interface and voice-enabled support could greatly enhance usability for non-English speakers, and broaden the system's reach worldwide.

Another promising avenue is to integrate patient metadata like age, skin type, medical history, and geographic location into a multi-input deep learning algorithm. This would improve diagnostic performance by offering the model more contextual information, lowering bias and achieving better performance on diverse populations.

Furthermore, using federated learning may enable the system to keep improving from decentralized data while protecting patient privacy, making it safer and more ethical. Finally, collaboration with dermatologists and healthcare organizations can be used to validate the system in hospitals, tune the prediction labels, and establish a feedback loop to ensure ongoing improvement.

Chapter 8

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

8.1 Alignment with SDGs

Explain how your project aligns with the United Nations Sustainable Development Goals. This can be a general overview referencing the SDGs that relate to your work. Example: The project aligns with several Sustainable Development Goals (SDGs), including: • SDG 9: Industry, Innovation, and Infrastructure – by leveraging advanced computing and innovation to solve practical problems. • SDG 4: Quality Education – if the project has an educational or accessibility focus, such as an AI-based learning system. • SDG 3: Good Health and Well-being – if the project is in the healthcare domain, such as disease prediction or mental health monitoring. • SDG 11: Sustainable Cities and Communities – if the project involves smart cities, urban mobility, or resource optimization.

8.2 Relevance of the Project to Specific SDG

This Project has the following Sustainable Development Goals (SDGs) directly supported by it. The main aim of this project is closely related to SDG 3: Good Health and Well-being, which focuses on ensuring healthy lives and promoting well-being for all at all ages. Early Detection Diagnosis: Early detection of skin diseases

such as carcinoma, eczema, and rosacea is possible through AI-based image classification, possibly preventing the advancement of severe conditions such as skin cancer.

Accessibility to Healthcare: In low-resource environments, particularly in rural or underserved communities, access to dermatologists is scarce. An AI-powered tool implemented through mobile or web apps can aid in primary skin disease screening, empowering community health workers and patients themselves.

Decrease in Healthcare Burden: Automating the initial triage and detection saves healthcare professionals from unnecessary burdens, allowing them to prioritize patients quickly and better.

This project further assists SDG 9: Industry, Innovation, and Infrastructure through the incorporation of innovative technology within the health sector.

Deep Learning and AI Use: Utilizing Convolutional Neural Networks (CNNs) for classifying medical images is a crucial technological leap for digital health diagnostics.

Scalable Deployment Potential: With the likes of Streamlit, cloud-based services, and mobile deployment, the solution is scalable, flexible, and cost-efficient as it is by nature for actual-world healthcare systems.

Bridging the Digital Divide: By utilizing lightweight and open-source frameworks, this project promotes democratization of health tech so that even small clinics or NGOs can implement AI-based solutions without huge investment.

The project has a direct applicability to SDG 3: Good Health and Well-being in improving early diagnosis as well as the propagation of accessible dermatological screening. It also aligns with SDG 9: Industry, Innovation, and Infrastructure through the incorporation of AI-based diagnostic tools that are scalable and cost-effective for wider use.

8.3 Potential Social and Environmental Impact

Social Impact – Corresponds to SDG 3:

Good Health and Well-being Enhanced Access: The project offers an AI-based di-

agnostic device that can be utilized in distant and underserved communities, particularly where dermatologists are not easily accessible. This enhances health equity by extending much-needed diagnostic support to low-resource settings. **Early Detection Intervention:** By facilitating early detection of skin conditions like carcinoma, eczema, and rosacea, the system facilitates timely intervention, slowing down disease progression, and enhancing long-term health results. **Affordable Scalable Solution:** By using open-source tools and lean deep learning models, the system can be cost-effectively rolled out across clinics, mobile health units, and telemedicine platforms—providing an inclusive healthcare solution. **Digital Health Education:** The system can also be used as an educational tool to train community health workers or medical students in detecting skin conditions via annotated predictions and model outputs.

Environmental Impact – Aligns with SDG 9:

Industry, Innovation, and Infrastructure Low-Energy Processing: The system is constructed to be run on comparatively light hardware (e.g., smartphones or micro-devices), minimizing the requirements for energy-intensive computations. This allows for energy saving and sustainability in digital health. **Reduced travel emissions:** Through remote screening capabilities, the system minimizes the requirement for patients to travel to visit specialists for far distances, hence minimizing carbon emissions from medical transportation. **Healthcare Resource Optimization:** Pre-screening automation enables healthcare systems to focus on critical cases first, resulting in better utilization of medical staff and resources, fewer unnecessary diagnostic tests. This initiative furthers SDG 3 (Good Health and Well-being) by improving health access, affordability, and outcomes through AI-supported dermatological diagnosis. At the same time, it supports SDG 9 (Industry, Innovation, and Infrastructure) by advancing eco-friendly, scalable, and energy-efficient digital health.

Chapter 9

PLAGIARISM REPORT



Figure 9.1: Plagiarism Report

Chapter 10

SOURCE CODE

10.1 Source Code

```
1 import streamlit as st
2 from PIL import Image
3 from keras.preprocessing.image import load_img, img_to_array
4 import numpy as np
5 from keras.models import load_model
6
7 model = load_model('./Model/BC.h5', compile=False)
8 lab = {0: 'Acne', 1: 'Carcinoma', 2: 'Eczema', 3: 'Keratosiis', 4: 'Milia', 5: 'Rosacea',}
9
10 def processed_img(img_path):
11     img=load_img(img_path,target_size=(224,224,3))
12     img=img_to_array(img)
13     img=img/255
14     img=np.expand_dims(img,[0])
15     answer=model.predict(img)
16     y_class = answer.argmax(axis=-1)
17     print(y_class)
18     y = " ".join(str(x) for x in y_class)
19     y = int(y)
20     res = lab[y]
21     print(res)
22     return res
23
24 def run():
25     img1 = Image.open('./meta/logo1.png')
26     img1 = img1.resize((350,350))
27     st.image(img1,use_column_width=False)
28     st.title("Skin conditions classification using cnn")
29     st.markdown(''''<h4 style='text-align: left; color: #d73b5c;'>* Data is based " six
        skin conditions datasets"</h4>''',
        unsafe_allow_html=True)
30
31
32     img_file = st.file_uploader("Choose an Image of Skin", type=["jpg", "png"])
```



```

33     if img_file is not None:
34         st.image(img_file, use_column_width=False)
35         save_image_path = './upload_images/'+img_file.name
36         with open(save_image_path, "wb") as f:
37             f.write(img_file.getbuffer())
38
39     if st.button("Predict"):
40         result = processed_img(save_image_path)
41         st.success("Predicted skin conditions is: "+result)
42 run()

```

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